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Q1.(Array Properties)

```
import numpy as np

arr = np.random.randint(10, 101, size=(5, 6))

print("2D Array:")

print(arr)

print("\nShape of the array:", arr.shape)

print("Total number of elements (size):", arr.size)

print("Data type (dtype):", arr.dtype)
```

Q2.Statistical Methods - Basic)

```
import numpy as np

arr = np.random.randint(1, 51, 20)

print("Array:")

print(arr)

print("\nMinimum Value:", arr.min())

print("Index of Minimum Value:", arr.argmin())

print("\nMaximum Value:", arr.max())

print("Index of Maximum Value:", arr.argmax())

print("\nSum of All Elements:", arr.sum())

print("Mean:", arr.mean())

print("Standard Deviation:", arr.std())
```

Q3.Statistical Methods on 2D Array)

```
import numpy as np

matrix = np.random.randint(20, 81, size=(4, 5))

print("Matrix:")

print(matrix)

print("\nMinimum Value:", matrix.min())

print("Maximum Value:", matrix.max())

print("Sum of All Elements:", matrix.sum())

print("Mean:", matrix.mean())

print("Standard Deviation:", matrix.std())

print("\nSum of Each Row:")

print(matrix.sum(axis=1))

print("\nSum of Each Column:")

print(matrix.sum(axis=0))
```

Q4.Reshape

```
import numpy as np

arr = np.arange(1, 25)

print("Original 1D Array:")

print(arr)

print("Shape:", arr.shape)
```

```
arr1 = arr.reshape(4, 6)
print("\nReshaped Array (4 x 6):")
print(arr1)
print("Shape:", arr1.shape)
arr2 = arr.reshape(3, 8)
print("\nReshaped Array (3 x 8):")
print(arr2)
print("Shape:", arr2.shape)
arr3 = arr.reshape(2, 3, 4)
print("\nReshaped Array (2 x 3 x 4):")
print(arr3)
print("Shape:", arr3.shape)
```

Q5.NumPy Indexing - 1D & 2D)

```
import numpy as np
arr = np.array([[10, 20, 30, 40],
                [50, 60, 70, 80],
                [90, 100, 110, 120]])
print("Original Array:")
print(arr)
print("\nFirst Row:")
print(arr[0])
print("\nLast Column:")
```

```
print(arr[:, -1])  
  
print("\nCenter 2 x 2 Submatrix:")  
  
print(arr[1:3, 1:3])  
  
print("\nEven Numbers:")  
  
print(arr[arr % 2 == 0])
```

Q6.Advanced Indexing)

```
import numpy as np  
  
arr = np.random.randint(1, 101, size=(5, 5))  
  
print("Original Array:")  
  
print(arr)  
  
print("\nDiagonal Elements:")  
  
print(np.diag(arr))  
  
print("\nElements Greater Than 50:")  
  
print(arr[arr > 50])  
  
arr[arr < 30] = 0  
  
print("\nModified Array:")  
  
print(arr)
```

Q7.Arithmetic Operations)

```
import numpy as np
```

```
a = np.array([10, 20, 30, 40])
b = np.array([1, 2, 3, 4])
print("Array a:", a)
print("Array b:", b)
print("\nAddition:")
print(a + b)
print("\nSubtraction:")
print(a - b)
print("\nMultiplication:")
print(a * b)
print("\nDivision:")
print(a / b)
print("\nElement-wise Power:")
print(a ** b)
print("\nModulo Operation:")
print(a % b)
```

Q8.Matrix Multiplication)

```
import numpy as np
A = np.array([[1, 2, 3],
              [4, 5, 6],
              [7, 8, 9]])
B = np.array([[9, 8, 7],
```

```

        [6, 5, 4],
        [3, 2, 1]])
print("Matrix A:")
print(A)
print("\nMatrix B:")
print(B)
element_mul = A * B
print("\nElement-wise Multiplication:")
print(element_mul)
matrix_mul = A @ B
print("\nMatrix Multiplication:")
print(matrix_mul)

```

Q9.Combined - Properties + Operations + Indexing)

```

import numpy as np
arr = np.random.randn(6, 6)
print("Original Matrix:")
print(arr)
print("\nShape:", arr.shape)
print("Size:", arr.size)
print("Data Type:", arr.dtype)
print("\nIndex of Maximum Value:", np.unravel_index(arr.argmax(), arr.shape))

```

```
print("Index of Minimum Value:", np.unravel_index(arr.argmin(), arr.shape))

print("\nTop-left 3 x 3 Submatrix:")

print(arr[:3, :3])

arr = np.abs(arr)

print("\nModified Matrix (Absolute Values):")

print(arr)

print("\nMean of Modified Matrix:", arr.mean())
```

Q10. Mini Project - Student Performance Analysis)

```
import numpy as np

marks = np.random.randint(30, 101, size=(10, 5))

print("Student Marks (10 x 5):")

print(marks)

total = marks.sum(axis=1)

average = marks.mean(axis=1)

print("\nTotal Marks of Each Student:")

print(total)

print("\nAverage Marks of Each Student:")

print(average)

highest = total.argmax()

lowest = total.argmin()
```



```
print("\nStudent with Highest Total Marks:")
print("Student Index:", highest)
print("Total Marks:", total[highest])
print("\nStudent with Lowest Total Marks:")
print("Student Index:", lowest)
print("Total Marks:", total[lowest])
print("\nOverall Class Mean:", marks.mean())
print("Overall Class Standard Deviation:", marks.std())
top3 = np.argsort(total)[-3:][::-1]
print("\nTop 3 Student Indexes:")
print(top3)
print("\nMarks of Top 3 Students:")
print(marks[top3])
reshaped = marks.reshape(5, 10)
print("\nReshaped Array (5 x 10):")
print(reshaped)
```